BATIMIX: response to the design critique

Peer Critique Response, Week 12 · DS5343 Data Visualization, UTEC 2026-2 · Jimena Gurbillon, Andrea Coa, Paolo Medrano · 28 October 2026 (updated 2 November 2026)

This document summarizes the feedback received in the Week 10 design review and records what we did with each substantive suggestion: **implemented**, **partial**, **postponed** or **rejected**, with its rationale and where it can be verified in the application (`deliveries/week12/app/`).

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suggestions recorded	implemented	partial	postponed	rejected

1. Feedback received

Instructor's remarks in the design review. There should be no white space between charts (no wasted space); the flow should be smooth; the interface should be friendly and free of visual noise: no overlapping text, and filters and interactive tools should not add noise.

Team decisions after the review. No scrollytelling or animations (motion graphics): the whole analysis must fit in a single view. One header with all the controls; remove the texts that explain each chart and how to use it; remove the space between visualizations; replace the toggle-based topomap with a 2 × 3 matrix (NS, SD, Δ by σθ and μθ, or by two people chosen in the scatter). After reviewing a first redesign, the team also asked for a complete visual change and for removing the noise in the multiverse panel.

Questions we asked the reviewers (`deliveries/week10/PrototypeReport.md`): six questions about the introduction, the lasso, map aggregation, the compact layout, how missing data are communicated, and the interpolated field. We answer them below with what changed.

Later literature review (`research/literatura_verificada.md` and `research/informe_conectividad.md`, in Spanish): methodological suggestions that arose while verifying the state of the art.

2. Response log

ID	Source	Suggestion	Decision	What we did and why	Where to see it
P1	Instructor	No white space between charts and no wasted space.	**Implemented**	Contiguous panels, separated only by 1-px lines, that fill the screen; each chart measures its panel and redraws when it changes. At 1365 × 768 the page does not scroll and no panel overflows.	Full view at 1365 × 768 and 1600 × 900
P2	Instructor	Smooth flow.	**Implemented**	Three columns with a clear hierarchy: the people on the left, the topomap matrix and the five-minute series in the center, secondary views on the right; each panel title is its finding, so the flow reads at a glance.	Whole view

ID	Source	Suggestion	Decision	What we did and why	Where to see it
P3	Instructor	Friendly interface.	**Implemented**	Titles that state the finding with its figures, a plain-language group selector and an *About* panel with introduction, glossary, tasks and method.	Panel titles; About button
P4	Instructor	No visual noise: no overlapping text; filters and tools without noise.	**Implemented**	All controls in one header; no role labels, chips or Play button; no electrode names on the maps (they are in the tooltip); separated counts and axes; axis titles shortened when they do not fit. Verified without overlaps at 1365 × 768 and 1600 × 900.	Header; all panels
E1	Team	No scrollytelling or motion graphics; everything in one view.	**Implemented**	The scrolling story and all data transitions were removed. Motivation, data and conclusion are in *About*; the findings, in the titles.	Whole app
E2	Team	One header; no explanatory text or space between visualizations.	**Implemented**	A single header with region, metric, group, marker, Clear and About. All usage instructions and the walkthrough column were removed.	Header
E3	Team	Six topomaps in a 2 × 3 matrix; rows variability and mean, or two participants chosen in the scatter.	**Implemented**	NS, SD and Δ columns; σθ and μθ rows of the active group. With exactly two people selected (click on points or heatmap columns), the rows become those two people with the header metric and shared scales. Δ is defined as SD minus NS throughout the app, so that positive means an increase after deprivation.	Center panel
E4	Team	Complete visual redesign (the first redesign was not convincing).	**Implemented**	A light classic theme with ColorBrewer palettes (default) and an optional dark "instrument" theme, both chosen by the team; NS blue, SD red, Δ diverging with a neutral zero.	Whole app; ◐ button
E5	Team	The multiverse panel looked cluttered and overlapping.	**Implemented**	The 19-row mark matrix and the 448 whiskers were replaced by a clean curve over the chance band and a dot plot with the median and interquartile range of r by decision. After the delivery, the team removed the panel to reduce the number of charts; the analysis remains in About (Robustness) and in the Report.	About
E6	Team	The way sex and other groups were shown did not work well.	**Implemented**	The single-group filter was replaced by *Compare*: two groups with their own colors in every view (map rows, ρ per group, Δ per epoch for each group).	Compare selector

ID	Source	Suggestion	Decision	What we did and why	Where to see it
E7	Team	A report tool to download and understand the selection.	**Implemented**	*Report* button: self-contained HTML with the state of the view, the charts, how to read each one, the people table and its CSV; it can be printed or saved as PDF.	Report button
E8	Team	Better, shorter titles for each task; everything interactive.	**Implemented**	A short headline computed from the data. Electrodes and time windows redefine the scatter x-axis; clicking points or heatmap columns builds the selection; all views are linked.	Panel titles
E9	Team	Add a graph and a subject × electrode heatmap, integrated in the same view, with draggable and groupable nodes.	**Implemented**	An *Individuals* floating panel that slid over the right column without leaving the view; after the delivery, the heatmap became a fixed panel and the graph was removed to reduce the number of charts. The heatmap uses two behavior-defined groups (not a third group without a marker, which answered no task) with their consistency C(e); selected people are outlined without reordering.	Heatmap panel
Q1	Our question	Does opening directly on the dashboard give enough context?	**Implemented**	It did not. The header now has the title and a pipeline chip, and *About* provides the introduction, data, findings and preliminary conclusion.	Header and About
Q2	Our question	Free lasso, rectangle, or multiple selection from the table?	**Partial**	The lasso is kept (it includes the rectangle), click selection from 1 to n people was added, and there is a keyboard-accessible selector of predefined groups. Group selection from the table remains for later.	Scatter; group selector
Q3	Our question	Is it clear that the map stays aggregated while other views follow the selection?	**Implemented**	The six maps are recomputed for the active group and the title says how many people they average; per-electrode significance is shown only for all 68 people.	Topomap matrix
Q4	Our question	Is the compact layout readable on a laptop?	**Implemented**	Tested at 1365 × 768, 1600 × 900 and 390 px wide; on narrow screens the panels stack, maps first.	Screenshots in `assets/`
Q5	Our question	Are missing data and coverage communicated well?	**Implemented**	Each title shows its effective n; About lists the paired n of every marker.	Titles; About

ID	Source	Suggestion	Decision	What we did and why	Where to see it
Q6	Our question	Keep the interpolated field or use discrete points?	**Partial**	Both are kept: a field interpolated with v4 (the method of EEGLAB *topoplot* and FieldTrip, verified against both in MATLAB) on the real electrode positions, with a selector that also shows MNE's cubic and the spherical spline (the v4 map correlates $r \geq 0.985$ with MNE's and $r \geq 0.93$ with the spline), and electrodes as points, with a ring ($p < 0.05$) and fill (FDR) in the Δ column. It follows the precedent of Bai et al. (2024).	Matrix, Δ column
L1	Literature	Median RT is not the best primary PVT measure (Basner and Dinges, 2011).	**Implemented**	PVT lapses and RT SD were added to the Y-axis and to the multiverse.	Y-axis; multiverse
L2	Literature	Explain the non-replication of Cui et al. (2026).	**Implemented**	A Multiverse with permutation inference, now 896 analyses with the preprocessing pipeline as a dimension: no joint statistic exceeds chance (σθ only: $p = 0.11$). The main pipeline now uses average reference and ICA, as Cui et al. and the sleep-deprivation literature, and an exact Cui-style rerun also gives no link with the PVT ($\rho = -0.06$). In addition, the «Oscillatory or 1/f?» view shows that σθ is not an artifact of the 1/f background.	About, Robustness; secondary views
L3	Literature	Rerun the preprocessing with average reference and ICA, as Cui et al.	**Postponed**	The raw EEG is now downloaded and the pipeline is verified to reproduce it (`tools/check_reproduction.py`), and the synchrony view already uses average reference. Rerunning σθ with average reference and ICA as a new multiverse dimension is scheduled for Week 13 (Jimena).	Plan, section 4

ID	Source	Suggestion	Decision	What we did and why	Where to see it
L4	Team and literature	Add a functional connectivity view (graph between electrodes).	**Implemented**	On 28 October we rejected it for the main board: the most similar study found no effects with eyes open, the pipeline used FCz reference and 1830 edges required large effects (decision matrix 29/40 versus 24/40). **Update (2 November):** the condition the report itself set for reopening it was met, because we downloaded the raw EEG and verified that the pipeline reproduces it. We added it as an exploratory secondary view with the safeguards from the literature: FCz re-inserted and average reference, debiased wPLI², 10 regions and 45 pairs, FDR and permutation, shown as a head graph and an effect-per-pair plot with 95% CIs. Result: theta synchrony decreases in all 45 pairs, but none survives FDR (joint p = 0.055) and it is unrelated to the PVT; it is labeled exploratory because at rest it may reflect aperiodic activity (Monchy et al.).	Right column, «θ synchrony between regions»; `tools/build_secondary.py`
U1	Rubric	Evaluate with users.	**Postponed**	Informal test with 3 to 5 users (tasks T1 to T8, think-aloud) in Week 13.	Plan, section 4

3. Peer cards (I like / I wish / What if)

Space to transcribe the cards written by classmates in the Week 10 review, with their responses in the same form as the table above. The instructor's suggestions (P1 to P4) are already recorded above.

4. Pending for the final three weeks

Week 13: pipeline with average reference and ICA (Jimena), moderators and teaser script (Andrea), vulnerability by lapses and Results (Paolo), deployment and user test (everyone). Week 14: paper in the IEEE VIS template, poster, videos. Week 15: final polishing, README, contributions, reflection and exhibition.